MRI Reconstruction for Knees Using Swin-Unet

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***Abstract*—Magnetic resonance imaging (MRI) is an essential modality in clinical diagnosis. However, its lengthy scanning time and susceptibility to motion artifacts caused by patient movement during prolonged imaging are significant drawbacks. These issues can result in patient discomfort and anxiety, making accelerated imaging indispensable for MRI. While convolutional neural network (CNN) based methods currently serve as the gold standard for medical image reconstruction, the limited ability of CNNs to capture long-distance information can lead to defects in the reconstructed images, such as blurred contours. In this project, we aim to address this limitation by harnessing the power of transformers in capturing long-distance information. We adopted a novel Swin-Transformer U-Net model which combines the U-Net and Swin Transformer. Training and evaluating on the MRNET dataset, our method achieved superior performance compared to the existing U-net based model in MRI reconstruction tasks, improving 5.8% in PSNR and 7.1% in SSIM, which provides a new approach to solve this challenging problem. Our experiments proved reconstruction in the image domain should be the most effective.**

***Index Terms*—MRI reconstruction, k-spaces, SwinUnet, transformer.**

I. Introduction

Magnetic Resonance Imaging (MRI) is a non-invasive medical imaging technique that reveals distinct anatomical structures. However, its lengthy data acquisition poses a significant challenge, limiting spatial and temporal resolutions and causing motion artifacts. This inefficiency hampers clinical applications requiring high-resolution imaging. To expedite MRI scans, three main approaches are employed: physics-based fast imaging sequences, hardware-based parallel imaging with multiple coils, and signal processing-based MR image reconstruction from incomplete k-space data. These methods, either alone or in combination, reduce acquisition and reconstruction times [1–3]. In this project, our primary focus is on signal processing-based techniques, which utilize prior knowledge to regularize the image reconstruction process from limited measurements.

In recent decades, MR image reconstruction has transitioned from linear analytic to nonlinear iterative methods. Analytic techniques like partial Fourier, SENSE [4], SMASH [5], and GRAPPA [6] optimize scanner efficiency using K-space correlations. Nonlinear methods, incorporating priors such as sparsity [7] and low-rank [8, 9], consider system physics and object knowledge. These methods, developed through model-based or manual approaches like dictionary learning, face challenges in prior model capacity and CS parameter tuning. The trend favors data-driven signal models over human-defined ones, addressing limitations in both linear and nonlinear approaches in commercial MR scanners.

Deep learning, a critical AI technology, utilizes layered neural networks to analyze intricate patterns in extensive datasets. In 2016, inspired by the success of deep learning, especially convolutional neural network (CNN), in computer vision (CV) and image processing tasks, Wang et al. pioneered CNN for MRI reconstruction. They introduced an offline CNN to learn the mapping between zero-filled MR images and fully sampled images, and resulted in superior performance [10]. In 2018, Zhu et al. further proposed the AUTOMAP network, a feed-forward network with convolution layers and fully connected layers [11]. Since then, various versions of neural network architectures were developed in order to enhance the speed and quality of reconstruction. Notable examples include U-net [12, 13], Residual Network (ResNet) [14, 15] and Generative Adversarial Network (GAN) [16, 17]. These studies, with promising performance achieved, highlight the great power and potential of deep learning in reshaping the rapid MR imaging techniques.

For MRI data, there are two domains: the image and frequency domains. The image domain represents the visualization of spatial MRI data, with pixels corresponding to specific anatomical locations. The MRI data is depicted as an intensity matrix, showing the strength of the signal. On the other hand, the frequency domain depicts the data as spatial frequencies, indicating variations in the signal across different frequencies. K-space serves as a mathematical representation capturing the frequency content of the image. During scanning, data is collected and processed to generate the final image. In MRI, the correlation between pixels in the frequency domain (k-space) and the image domain is established by applying Fourier transformation. Therefore, fast MR imaging research can be conducted based on either the frequency (k-space) or image domain. Currently, although there are several models dedicated to optimizing both domains [18–20], due to the interrelationship between these two domains, most end-to-end deep learning MRI studies begin from the image domain, leveraging techniques such as GAN and U-Net, while a smaller portion start from the k-space domain, for instance with Automated Transform by Manifold Approximation (AUTOMAP).

In addition to the correlation between frequency and image domains, it is evident, based on the principles of MRI generation, that each pixel within each domain is inherently correlated with one another. Let's consider the image domain as an example. In MRI, the correlation of pixels in the image domain is a consequence of the underlying physical phenomena being measured. The interaction between tissue magnetic properties and applied magnetic fields results in a spatially complex signal distribution. During Fourier transformation processing to generate the image, each pixel represents a specific object feature, leading

to correlations between neighboring pixels due to shared tissue properties or structures. The Fourier transform converts spatial data into frequencies, linking spatial variations to frequency components. This correlation in MRI images reflects the intrinsic spatial relationships within the imaged object, which is crucial for accurately depicting anatomical details.

Utilizing the inherent correlation among pixels is crucial for accurate image reconstruction. As the classic CNN is constrained by the limited size of the convolution receptive field, it is unable to capture long-range dependencies in images, resulting in inaccurate reconstruction of image details and textures. Drawing inspiration from the exceptional performance of the Transformer model in natural language processing (NLP), researchers have integrated Transformer architecture into the field of CV. Transformers are neural network architectures designed for processing sequential data such as text or time series. Typically, a transformer is composed of an encoder and decoder with layers of self-attention and feedforward networks. Unlike recurrent neural networks (RNNs) or CNNs, transformers rely on self-attention mechanisms to effectively capture long-range dependencies in data, making them well-suited for contextual understanding [21]. The aforementioned advantages may be utilized to exploit the long-range relationships among pixels in MRI reconstruction tasks, as evidenced by the superior reconstruction performance achieved in several recent studies [22–25].

In this project, we examined MR reconstruction using Swin-Unet in the image and frequency domain, or dual domain. These experiments help us understand better how SwinGAN works on reconstruction. The Swin-Unet introduces transformer structures to learn correlation among each pixel. We also extended the reconstruction from the single-channel amplitude representation of MR images to directly reconstructing the full complex form of images, which is derived from the inverse Fourier transform of k-spaces. Our experiments in the image domain achieved satisfactory outcomes, demonstrating that the transformer structure significantly benefits image domain reconstruction. Our work could be great reference for further research of fast MRI reconstruction.

II. Swin-Transformer and Swin-Unet

Due to the limited extent of the convolutional receptive field, networks based on CNNs often face challenges in capturing long-range dependencies in images. This limitation can lead to discrepancies in accurately reconstructing intricate image details and textures. Inspired by the pioneer application of the Transformer applied in the NLP fields, researchers have brought the Vision Transformer (ViT) into the CV field. The ViT employs a Transformer architecture as a feature extractor for image construction. It processes 2D image patches as input and undergoes pretraining on extensive dataset such as ImageNet, achieving comparable or even superior performance compared to CNNs. However, the original transformer ignored the local contextual information in the input data, so we tried to explore a model that can keep the relative position of the image patches.

We mainly used Swin-Unet as our backbones in several experiments. The Swin-Transformer U-Net architecture combines the principle of U-Net architecture and Swin Transformer, leveraging the strengths of convolutional neural networks and self-attention mechanisms. Figure 1 shows the architecture of our Swin-Transformer U-Net. It consists of encoder, bottleneck, decoder, and skip connections.

In the encoder, the initial image is partitioned into non-overlapping patches, mirroring the patch-based method in ViT. Subsequently, each patch undergoes a linear projection into a lower-dimensional space(embedding), facilitating the capture of pixel spatial relationships within the patch. The transformed patches traverse several Swin-Transformer blocks and patch merging layers to generate hierarchical feature representations.

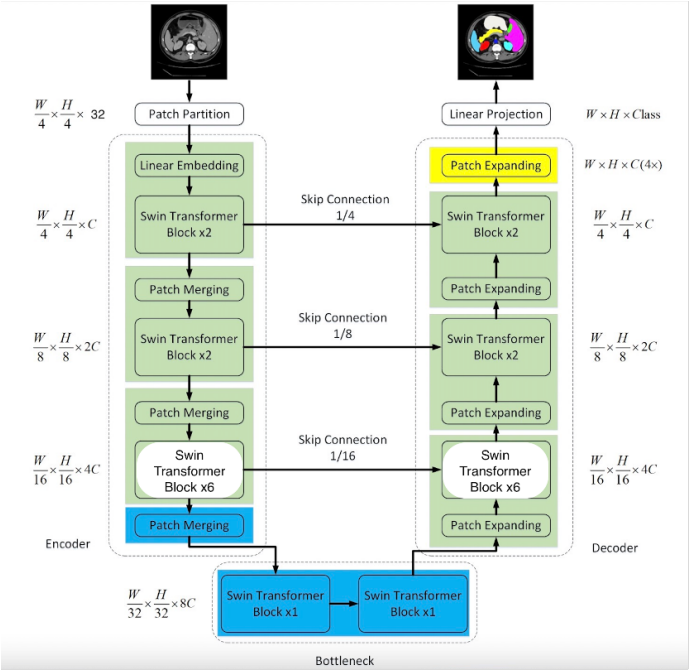

Figure 1: Swin-Unet Architecture. This figure is mainly from Swin-Unet [26]

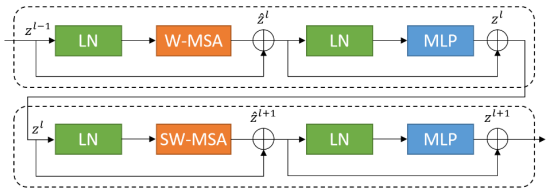

Figure 2: Swin-Transformer Block. This figure is from Swin-Unet [26]

In the bottleneck, two consecutive transformer blocks are applied, which are the window-based multi-head self attention (W-MSA) module and the shifted window-based multi-head self attention (SW-MSA) module. Each block consists of LayerNorm (LN) layer, multi-head self attention module, residual connection and 2-layer MLP with GELU [27].

In the decoder, there are Swin Transformer blocks and patch expanding layers. The patch merging layer is responsible for down-sampling and the patch expanding layer is responsible for up-sampling. The contexture features extracted are integrated with multiscale features from the encoder via skip connections

to complement the loss of spatial information caused by down-sampling. A final linear projection layer brings the up-sampled features to output the predictions.

The attention mechanism is similar to the one in transformer in NLP. It is specifically designed to compute interactions between elements within a limited window, thereby allowing the model to capture both local and global context. The key innovation of Swin-Transformer is the shift window mechanism, which re-organises the embeddings within each patch. It enables the model to capture the contextual details efficiently, hence acquiring long-range dependencies and pixel relationships across the entire image. The shifted window mechanism enhances efficiency by restricting self-attention calculations to distinct local windows without overlap, while also enabling inter-window connections. The Swin-Transformer forms a hierarchical representation by commencing with small-sized patches and progressively merging adjacent patches in deeper transformer layers. The computation complexity is linear to image size. Another crucial design aspect involves shifting the window partition between successive self-attention layers, as depicted in [27]. These shifted windows connect the windows from the previous layer, establishing connections among them that substantially enhance modelling capabilities.

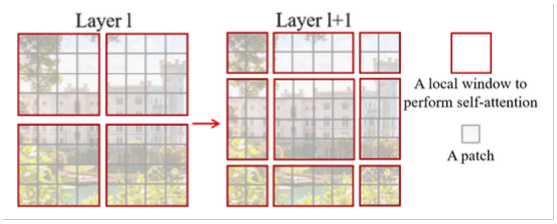

Figure 3: Shifting window. This figure is from Swin-Transformer [27]

We build our backbone Swin-Unet with embedding size C = 96, and depth for layer numbers = {2, 2, 6, 2}. The input is divided into non-overlapping patches with patch size of 4×4, resulting in the feature dimension of each batch to $4 \times 4 \times 2 = 32$. A linear embedding layer is applied on this feature to project it to dimension (denoted as C). The first patch merging layer combines the features of adjacent 2×2 patch groups and applies a linear transformation to the resulting concatenated features, reducing the number of tokens by a factor of 4 (representing a 2x down-sampling of resolution). The output dimension is adjusted to 2C. Subsequently, Swin Transformer blocks are utilized for future feature processing, maintaining the resolution at $\frac{H}{8}$ x $\frac{H}{8}$. The process repeated to result in $\frac{H}{16}$ x $\frac{H}{16}$ and $\frac{H}{32}$ x $\frac{H}{32}$. The decoder is performing patch expanding with dimension scale of 2 and Swin transformer block in a layer up fashion, similar to the structure of decoder steps.

It is such an improvement from ViT to Swin Transformer since Swin Transformer integrates the shift window mechanism, which rearranges embedding within patches to capture global context, therefore mitigating the limitation of sequential processing of patches by ViT. The shifted window mechanism presents numerous benefits compared to traditional sliding window techniques. It not only reduces the computational complexity by enabling parallelization across patches, but also is more flexible and adaptive compared to fixed-size sliding window, enabling the model to adapt various spatial contexts within the image.

III. Methods and Experiments

As described in previous sections, MR reconstruction encompasses two interconnected domain: the frequency and image domain, that could be converted to each other via Fourier transform or inverse Fourier transform. Then neural networks could be effectively employed to enhance reconstruction from under-sampled inputs , which may include under-sampled k-spaces or images in these two domains. We conduct experiments using Swin-Unet to reconstruct MR images from the under-sampled k-spaces and images respectively in the two domains. Additionally, we explored dual domain processing by combining the two Swin-Unet models, one for k-space and the other for the image domain, similar to the approach used in Swin-GAN [25]. The paper claimed that this paralleled architecture across two domains could not only maintain the consistency within each domain but also enhance the performance of reconstruction results. In the end, we could compare our results with the baseline using the classical Unet architecture without transformer structures.

We primarily trained and tested our models on MRNet dataset [28] from Stanford which includes coronal MR images of knees from 1,130 patients in the training dataset and 120 in the testing dataset. Note that MR reconstruction from given k-spaces should be 2D, and each patient's MR data consists multiple slices. For our study, we only retrieved the 10 middle slices from each patient for both the training and testing datasets. Consequently, the training datasets comprised 11,300 cases while the testing dataset consisted 1,200 cases. We could derive the fully sampled k-spaces from given fully sampled MR images by applying the inverse Fourier transform. We used the provided random masks of 20% sampling rate from Swin-GAN to multiply with fully sampled k-spaces to simulate under-sampled k-spaces. Random Gaussian noises were added as data augmentation. We could then got the under-sampled MR images by applying the Fourier transform on the under-sampled k-spaces.

We used Structure Similarity Index Measure (SSIM) and Peak-Signal-to-Noise Ratio (PSNR) to evaluate our reconstructed results. Higher SSIM and PSNR indicated better results.

(1) PSNR is a measure of the peak error between the original and the reconstructed image.

$$MSE = \frac{1}{N}\sum_{i=1}^{N}(I_y(i) - \hat{I}_y(i))^2 \tag{1}$$

$$PSNR = 10 \cdot \log_{10}\left(\frac{L^2}{MSE}\right) \tag{2}$$

where I_y denotes the real image, $\hat{I}_y$ denotes the reconstructed image, L refers to the max value of the image pixel, and N represents the number of pixels in the two images.

(2) SSIM could measure the perceived quality of reconstructed images by structural information, luminance, and contrast rather than the absolute pixel values.

$$SSIM(I, \hat{I}) = \frac{(2\mu_I\mu_{\hat{I}} + C_1)(2\sigma_{I\hat{I}} + C_2)}{(\mu_I^2 + \mu_{\hat{I}}^2 + C_1)(\sigma_I^2 + \sigma_{\hat{I}}^2 + C_2)} \tag{3}$$

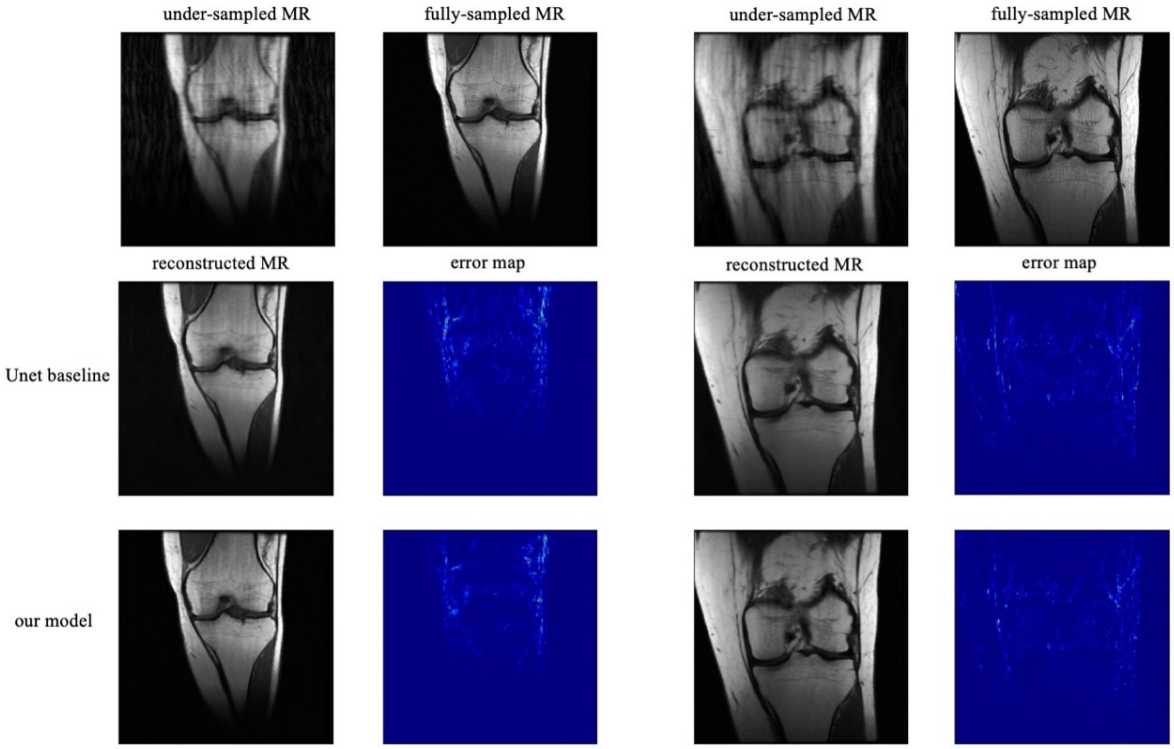

Figure 4: Reconstructed results comparison.

where I and $\hat{I}$ denote the ground truth and reconstructed MR image, μ and σ_I^2 are the mean and variance, $\sigma_{I\hat{I}}$ is the error between I and $\hat{I}$, and C_1 and C_2 are constant relaxation terms.

A. Reconstruction in the frequency domain

In this experiment, our model functioned as:

$$\begin{aligned} y &= \hat{y} + model\left(\hat{y}\right) \\ x &= \mathcal{F}^{-1}\left(y\right) \end{aligned} \tag{4}$$

where $\hat{y}$ represents the under-sampled k-spaces while y represents reconstructed k-spaces, and x is the reconstructed MR image derived by the inverse Fourier transform. We applied L1 loss both in frequency and image domains as below:

$$L = |y - \overline{y}| + |x - \overline{x}| \tag{5}$$

where $\overline{x}$ represents the fully sampled MR image as ground truth, and $\overline{y}$ represents the fully sampled k-space as ground truth. Using L1 loss in the image domain could help the model maintain reconstruction consistency in that domain and act as a regularization constraint. Notably, both the Fourier transform and the inverse Fourier transform are differentiable, allowing it to compute the gradient of the model effectively.

B. Reconstruction in the image domain

In this experiment, our model functioned as

$$\begin{aligned} \hat{x} &= \mathcal{F}\left(\hat{y}\right) \\ x &= \hat{x} + model\left(\hat{x}\right) \end{aligned} \tag{6}$$

where $\hat{x}$ represents the under-sampled MR image derived by the Fourier transform from under-sampled k-space $\hat{y}$, and x is the reconstructed MR image. Like previous experiment in frequency domain, we still applied L1 loss in the image domain and frequency domain while the L1 loss in the frequency domain served as the regularization term this time:

$$L = |x - \overline{x}| + |y - \overline{y}| \tag{7}$$

This experiment could be viewed as a mirrored counterpart, transitioning reconstruction from the frequency domain to the image domain.

C. Reconstruction in dual domains

In this experiment, our model consisted of two Swin-Unet models, one for the k-space domain and one for the image domain. It functioned as

$$\begin{aligned} y &= \hat{y} + freq_model\left(\hat{y}\right) \\ \hat{x} &= \mathcal{F}^{-1}\left(y\right) \\ x &= \hat{x} + image_model\left(\hat{x}\right) \end{aligned} \tag{8}$$

where $\hat{y}$ represents the under-sampled k-space, y represents the reconstructed k-space, $\hat{x}$ represents coarse reconstructed MR image, x is the refined MR image. The loss function was still L1 loss in two domains described below:

$$L = |y - \overline{y}| + |\hat{x} - \overline{x}| + |x - \overline{x}| + |\mathcal{F}\left(x\right) - \overline{y}| \tag{9}$$

where $\overline{x}$ represents the fully sampled MR image as ground truth, and $\overline{y}$ represents the fully sampled k-space as ground truth. Note that in each domain there is a pair of L1 loss in the two domains. $\hat{x}$ could be viewed as coarse Mr reconstructed images that could be fed into a second Swin-Unet in image domain for refinement. The refined MR images could also derive corresponding k-spaces to compute L1 loss regularization in the frequency domain.

D. Additional trials

We also attempted to directly model the reconstruction from k-spaces to MR images using Swin-Unet. However, this approach failed to converge and resulted in significantly inferior outcomes. Additionally, we incorporated adversarial learning like into our model for image domain reconstruction to see whether could improve the performance, similar to the approach taken by Swin-GAN, utilizing a discriminator to differentiate between the reconstructed MR images and the given ground truth images. Ideally, well reconstructed MR images should be able to deceive the discriminator. However, we observed a slight decline in performance compared to the model without adversarial learning., The GAN-like training introduced instability and failed to yield any improvements. As a result, we gave up adversarial training.

IV. Results

From the three experiments conducted on MR image reconstruction across image, frequency, and dual domains, we observed that the image domain reconstruction exhibited superior performance. The quantitative evaluation on the validation dataset is detailed in Table 1. From the table, we could draw the conclusion that the reconstruction solely within the frequency domain proved to be ineffective. Although the dual domain approach showed less impressive outcomes, these findings reinforce the dominance of image domain reconstruction in achieving superior reconstructed image quality. We also conducted reconstruction in image domain using a classical Unet architecture, the performance of which is presented in the table to serve as a baseline for comparison.

Table 1: Metrics for the results

Experiment/metrics	PSNR	SSIM
Image domain	30.8 $\pm$2.1	0.9 $\pm$0.03
Frequency domain	26.7 $\pm$1.6	0.76 $\pm$0.04
Dual domains	30.2 $\pm$1.8	0.87 $\pm$0.03
Baseline (Unet)	29.1 $\pm$1.7	0.84 $\pm$0.04

Figure 5 showcases the robust performance achieved by employing Swin-Unet for image domain reconstruction. Figure 6 shows that the L1 loss in image domain and L1 loss regularization constraint in frequency domain decreases with epoch during the training process. It indicates that our model is learning effectively without signs of overfitting. In addition, Figure 4 presented reconstructed results of our model in the image domain compared to the baseline using Unet. What's more, Figure 4 depicts the reconstruction results of two cases. It can be seen that our model using Swin-Unet reconstructed better textures and had less aliasing artifacts. Obviously, our model yield more satisfactory reconstruction of larger PSNR and SSIM.

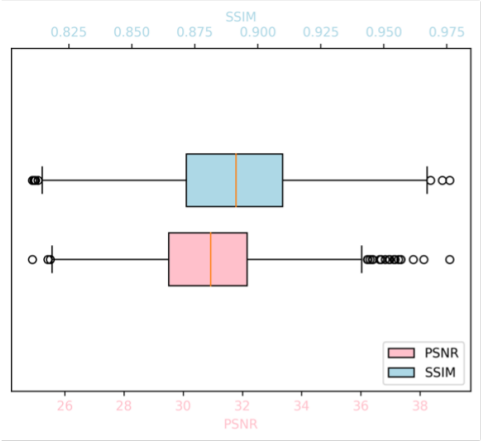

Figure 5: Performance of our model.

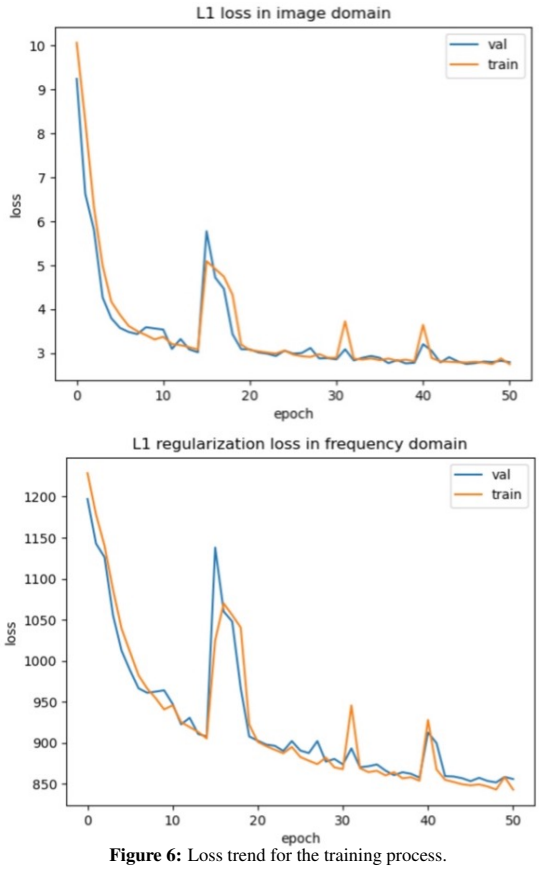

Figure 6: Loss trend for the training process.

V. Discussion

Our investigations into MR reconstruction across domains revealed that learning the residue between under-sampled and fully-sampled MR images in the image domain is effective, whereas a similar approach within the k-space did not yield successful outcomes. Therefore, we contend it is the Swin-Unet that focus on image domain reconstruction that plays the crucial rule in the functionality of Swin-GAN. The implementation of adversarial learning offered minimal enhancements to the results. These extra design complexity in Swin-GAN might be unnecessary. A solitary Swin-Unet, dedicated to image domain reconstruction, may suffice for effective reconstruction. Meanwhile, the L1 loss

of corresponding k-spaces between reconstructed results and the ground truth could enforce consistency within the frequency domain.

By incorporating Swin-Transformer modules into the Unet architecture, Swin-Unet excels at capturing global contextual information more effectively than conventional CNN-based methods. Swin-Unet addresses the limitation of short-distance connection in feature extraction encountered by CNN-based methods. With every pixel interconnected in both the image and frequency domains, the utilization of shifted windows and image patches facilitates a more thorough exploration of correlations, resulting in superior reconstruction outcomes.

Reconstruction in the frequency domain could be challenging, particularly because the un-sampled regions appear entirely blank in k-spaces, and each pixel in the k-space has a global impact on every pixel in the reconstructed images. A significantly error in one pixel of k-space can propagate throughout the reconstructed image, leading to noticeable discrepancies across all pixels. However, the under-sampled images contain information that is not fully resolved at every pixel, which can help mitigate reconstruction challenges. One of our innovations lies in our model's design to handle the reconstruction of MR images in their complex form, directly derived from the Fourier transform of under-sampled k-spaces. The MR images typically presented to us primarily represent the amplitude of these complex images, potentially discarding crucial phase information. In contrast to many existing models that focus solely on reconstructing from the amplitude of the complex images, our approach harnesses additional phase information to enhance the reconstruction process.

VI. Conclusion

Our project successfully demonstrated the effectiveness of using Swin-Unet for MR reconstruction in the image domain. The novel Swin-Unet model leveraged the strengths of transformer and a shifting window mechanism, capturing both gobal and local contextual information effectively. We also innovatively proposed a method to handle the reconstruction by learning the residuals in the complex form of undersampled MR images, thereby preserving additional phase information for more accurate restoration.

Our experiments thoroughly analyzed each component of Swin-GAN, concluding that the Swin-Unet operated in the image domain, is the most critical for successful reconstruction. The other components were found to be superfluous.

Our work underscored the significance of domain-specific adaptations and detailed the rationale behind MR reconstruction techniques. It has laid a foundation for future advancements in learning pixel correlations within the complex forms of undersampled images. Additionally, this project serves as a testament to the versatility and adaptability of transformer models, which, though originally developed for NLP tasks, effectively address complex challenges in other fields such as computer vision and medical image analysis.

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Table 2: Contributions of Team Members

Student	**Contributed Aspects**	**Details**
Tengya Peng	Framework, model training, data Processing, report writing	He conducted extensive preliminary research, designed the overall framework and approach for this project, and was responsible for constructing the network model and pipeline. He primarily authored the experiments, results, and discussion sections of the report.
Hao Zhang	GPU set up, baseline model implementation, data analyses and report writing	He set up the GPU for training the model, and worked on the baseline model implementation, data analyses and testing. He also tested the developed model and explored the hyperparameter tuning in Google colab. He primarily organized the discussion meetings and wrote meeting minutes for follow-up tasks. For the report, he wrote the abstract and introduction part.
Huan Zheng	Experiment running, parameter tuning, architecture report section writing	Adapted the code and config the environment to run on Google Colab with cuda. Compared results of evaluation metrics with different epochs, loss weight, and batch size and performed analysis on them. Investigated how to enhance GAN's performance in the model, making it more unsupervised. Responsible for architecture sections report writing. Coordinated team members to keep track of requirements.
Yuwen Wu	Analysing Official PyTorch implementation of SwinGAN method, conclusion part reporting writing	Researched several MRI Reconstruction baseline including some MRI classification methods.